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Update $f_m(x) = f_{m-1}(x) + \sum_{j=1}^{J_m} \gamma_{jm} I(x \in R_{jm})$

Step 3: output $\hat{f}(x) = f_m(x)$

Gradient Boosting for Classification

All the process are like the regression, but only difference is the loss fn.

i) Regression = m.s.e

ii) Classification = log loss

1st model.

Stage 1 $\rightarrow \log(\text{odds})$

$= \log\left(\frac{\text{total no.s of 1}}{\text{total no.s of 0}}\right)$

first model output.

use loge

~~not log10~~

gives ~~probability~~ a value. $\rightarrow 0.51$

~~0.51~~

Then probability :

$\frac{1}{1 + e^{-\log(\text{odds})}}$

prob 1.

$\rightarrow 0.625$



Cal error of model 1
 Stage 2: send error of model 1 to another.

$$\text{Pseudo residual} = y_i - \text{prob1.}$$

$$\text{If } 0, \quad 0 - 0.625 = -0.625$$

$$\text{If } 1, \quad 1 - 0.625 = 0.375$$

res1.

Stage 3: send to another. High bias
 Decision regression tree $\rightarrow$ weak learners.
 low depth.

cgpa	iq	res1

gives value of terminal regions.

For all leaf node calc leaf nodes:

Formula:

$\sum \text{Residual}$

$$\sum [\text{Previous Prob} * (1 - \text{Previous Prob})]$$

Colⁿ

pre-2 (probability) : $\sum \log(\text{odds})$ of that
 leaf node.

~~For res2:~~
~~Probability~~

$$\frac{1}{1 + e^{-2 \log(\text{odds})}}$$

$$\text{Then, res 2} = y_i - [f_0(n) + f_1(n)]$$